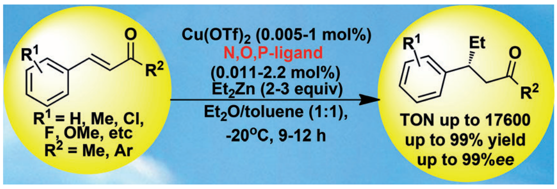
FULL PAPER



Match maker: The chiral bimetallic multinuclear Cu−Zn complex that contains a novel phosphine ligand and bears matching axial and sp^3-central chirality from binaphthol and chiral amine was found to be a highly efficient catalyst in the enantioselective conjugate addition of an organic zinc reagent.

Asymmetric Catalysis

*Fei Ye, Zhan-Jiang Zheng, Wen-Hui Deng, Long-Sheng Zheng, Yuan Deng, Chun-Gu Xia, Li-Wen Xu** ▮▮▮▮–▮▮▮▮

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Modulation of Multifunctional N,O,P Ligands for Enantioselective Copper-Catalyzed Conjugate Addition of Diethylzinc and Trapping of the Zinc Enolate

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Abstract: In this work, we have successfully synthesized a new family of chiral Schiff base–phosphine ligands derived from chiral binaphthol (BINOL) and chiral primary amine. The controllable synthesis of a novel hexadentate and tetradentate N,O,P ligand that contains both axial and sp^3-central chirality from axial BINOL and sp^3-central primary amine led to the establishment of an efficient multifunctional N,O,P ligand for copper-catalyzed conjugate addition of an organozinc reagent. In the asymmetric conjugate reaction of organozinc reagents to enones, the polymer-like bimetallic multinuclear Cu−Zn complex constructed in situ was found to be substrate-selective and a highly excellent catalyst for diethylzinc reagents in terms of enantioselectivity (up to >99% *ee*). More importantly, the chirality matching between different chiral sources, C_2-axial binaphthol and sp^3-central chiral phosphine, was crucial to the enantioselective induction in this reaction. The experimental results indicated that our chiral ligand (*R*,*S*,*S*)-**L1**- and (*R*,*S*)-**L4**-based bimetallic complex catalyst system exhibited the highest catalytic performance to date in terms of enantioselectivity and conversion even in the presence of 0.005 mol% of catalyst (S/C = 20000, turnover number (TON) = 17600). We also studied the tandem silylation or acylation of enantiomerically enriched zinc enolates that formed in situ from copper-**L4**-complex-catalyzed conjugate addition, which resulted in the high-yield synthesis of chiral silyl enol ethers and enoacetates, respectively. Furthermore, the specialized structure of the present multifunctional N,O,P ligand **L1** or **L4**, and the corresponding mechanistic study of the copper catalyst system were investigated by ^{31}P NMR spectroscopy, circular dichroism (CD), and UV/Vis absorption.

Keywords: asymmetric catalysis • conjugate addition • copper • N ligands • zinc

Introduction

Inherent in many metal-based asymmetric catalyst systems is the design and synthesis of versatile chiral ligands because it has a pivotal influence on catalytic activity and its ability to impact stereocontrol.[1] The art of designing these chiral ligands still remains quite empirical for various enantioselective transformations. In this context, a way to simplify the design of a consummate chiral ligand is to rely on modular structures that can take full advantage of steric and electronic properties of chiral molecules.[2] In other words, subtle modification and linking of modular ligands can easily lead to the dramatic formation of highly efficient ligands with excellent enantioselectivity and reactivity.[3]

Among various chiral ligands, multifunctional chiral ligands are probably the most interesting chiral scaffold used to promote enantioselective inductions in many enantioselective transformations and have been used to assemble different metals with a coordinate functional group to guarantee the achievement of high enantioselectivity and catalytic efficiency.[4] C_2-axial binaphthol (BINOL) derivatives are often used as chiral sources for multinuclear metal-based catalysts and offer promising possibilities for modification and derivatives,[5] which have expanded the scope of classic homogeneous catalysis to cooperative catalysis that arises from the synergistic and cooperative activation of natural enzyme biocatalysis.[6] In this regard, Endo and Shibata et al. have recently reported that a BINOL-based tetradentate phosphine ligand was of crucial importance for a bimetallic copper–zinc catalyst system in the activation of the carbonyl of chalcones, in which the catalytic performance achieved was much higher than that of previous systems in the copper-catalyzed asymmetric conjugate addition of organozinc reagents to chalcones.[7] Although enormous research

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efforts have been devoted to the synthesis of multinuclear metal catalysts, the development of new chiral ligands with excellent enantioselectivity in catalytic asymmetric reactions remains a challenge. Therefore, the design and synthesis of multifunctional chiral ligands associated with the construction of a multinuclear bimetal catalytic system for asymmetric reactions is highly desired.[8]

The question of how to develop a novel and effective chiral phosphine ligand with multistereogenic centers prompted us to investigate the modulation of C_2-axial and sp^3-central chirality in the development of highly efficient and multifunctional catalyst systems for enantioselective transformations. Considering the highly successful C_2-axial binaphthol (BINOL) and 2,2′-bis(diphenylphosphino)-1,1′-binaphthyl (BINAP) ligand in asymmetric catalysis, we hypothesized that a BINOL-derived N,O,P ligand that bears a hydroxyl and chiral amino moiety would be an ideal chiral ligand that bears axial and sp^3-central chirality to coordinate with different metals to form multinuclear transition-metal catalysts (Figure 1). Herein, based on the hypothesis that

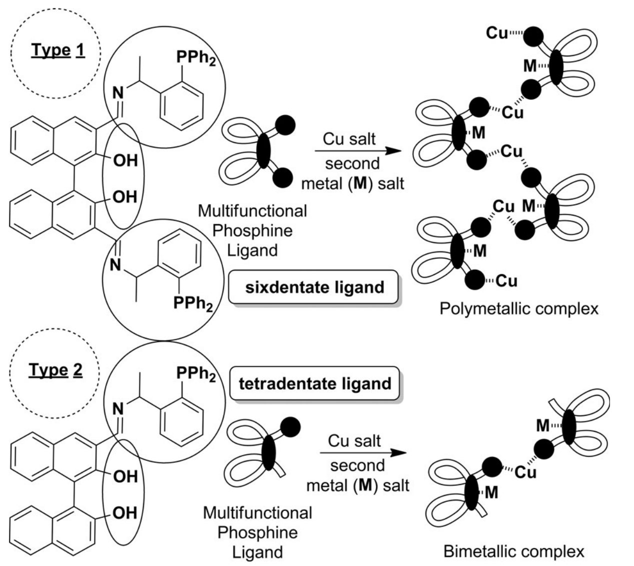


Figure 1. Working hypothesis for the construction of multinuclear polymetallic or bimetallic complexes: Which is better for catalytic asymmetric reactions?

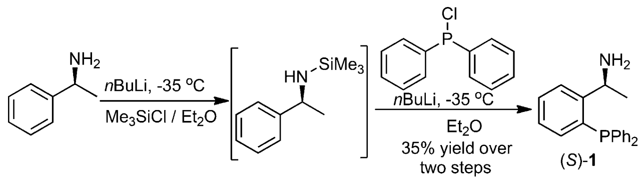
the introduction of an sp^3-central chiral phosphine to the chiral BINOL backbone is perhaps effective in the construction of a modular multinuclear polymetallic metal complex (type 1) or bimetallic metal catalyst system (type 2) for catalytic asymmetric reactions (Figure 1), we focused on the design and synthesis of a new family of BINOL-derived phosphine that bears two types of chiral elements, axial and sp^3-central chirality, and we demonstrate the different catalytic activity of these chiral N,O,P ligands when copper-catalyzed conjugate addition of organozinc reagents to enones is used as a model reaction.

Results and Discussion

Synthesis of Multidentate and Multifunctional Phosphine Ligands

Initially, we designed a hexadentate N,O,P ligand (Figure 1, type 1) and tetradentate N,O,P ligand (Figure 1, type 2) with different stereochemistry; they contained a tunable chiral environment and space around the 3,3′-position of BINOL (Figure 1) in which one or two imine phosphine molecules ((*S*)-1-[2-(diphenylphosphino)phenyl]ethanamine (**1**)) were linked to BINOL. The assumed multifunctional phosphine ligand (**L1**) and its variations in terms of structural diversity and the C_2-axial and sp^3-central chirality also needed to be used for the next catalytic transformation.

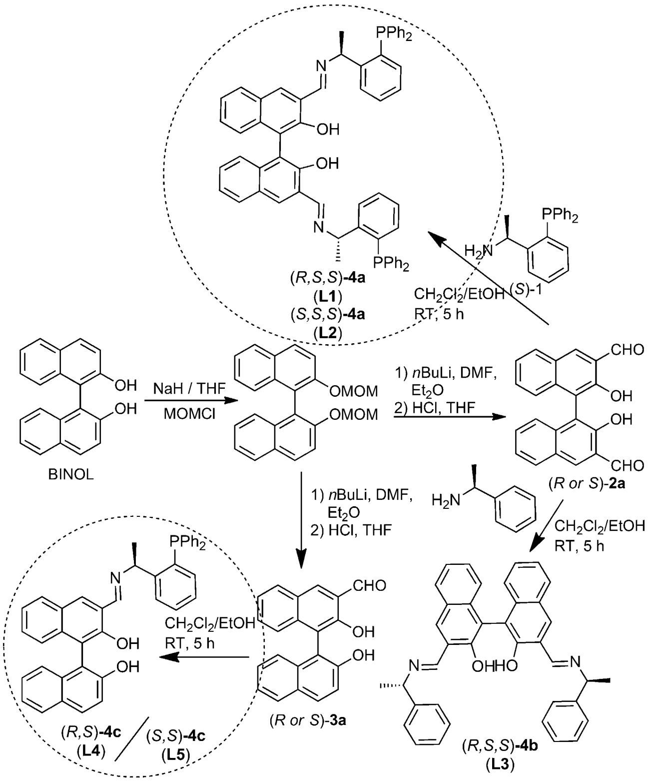
Compound **1**, which contained a chiral primary amine motif as a building block for the hexadentate ligand, was easily prepared from (*S*)-1-phenylethanamine in moderate yield (Scheme 1).[9] Simultaneously, two different chiral alde-



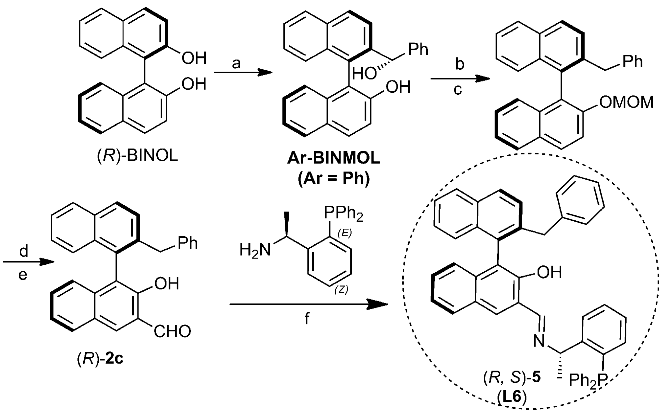
Scheme 1. Synthesis of chiral primary-amine-based phosphine as a building block.

hydes, 3-hydroxy-4-(2-hydroxynaphthalen-1-yl)-2-naphthaldehyde and bi-2-hydroxy-3-naphthaldehyde, were prepared from chiral BINOL in high yields (Scheme 2).[10] Then the five similar ligands (**L1**–**L5**) that contained both axial and sp^3-central chirality were prepared by Schiff base condensation of chiral primary-amine-based phosphine ligand **1** and different BINOL-derived aldehydes (**2a** or **3a**, 50–62 % total isolated yield; also see the Supporting Information).[10,11] Except for **L3**, these new multifunctional ligands featured three different coordinating sites: nitrogen, phosphine, and bidentate hydroxyl moiety. Notably, to highlight the importance of the phosphine center and BINOL backbone, the Ar-BINMOL-derived phosphine ligand **L6** was also prepared for the next asymmetric conjugate addition reaction of organozinc reagents to enone (Scheme 3).[12]

The crystal structure of the chiral ligand **L1** is shown in Figure 2. The phosphine ligand has a large cavity and the phosphine atoms extend outward due to steric hindrance of two phenyl groups. This beautiful structure would be a powerful chiral phosphine ligand to achieve high efficiency and enantioselectivity in asymmetric conjugate addition of organozinc reagents to enones, which is one of the most useful techniques for carbon–carbon formation in organic synthesis.[14] Considerable progress in asymmetric copper-catalyzed conjugate addition of dialkyl zinc reagents has been made and has generated tremendous interest in past decades.[15] In addition, the reaction of the organozinc reagent is an illustrative and model example of asymmetric multimetallic cat-



Scheme 2. Synthesis of various multidentate ligands containing phosphine and phenol groups with both axial and central chirality (MOMCl = methyl chloromethyl ether).



Scheme 3. Synthesis of chiral Ph-BINMOL-based aldehyde **2c**[10b] and corresponding phosphine ligand **L6**: a) [1,2]-Wittig rearrangement,[12] *i*BuLi, THF, −78 °C; b) TMSCl (TMS = trimethylsilyl), NaI; c) NaH, MOMCl; d) *n*BuLi, DMF, Et_2O; e) HCl, THF; and f) CH_2Cl_2/EtOH, at room temperature.

alysis.[13] Thus these observations inspired us with the confidence to study the catalytic activity of the new binaphthyl-modified phosphine and its analogues (**L1**–**L6**) as well as the chiral primary-amine-based phosphine ligand (*S*)-**1** in the model and enantioselective copper-catalyzed asymmetric conjugate addition of organozinc reagents to enones.

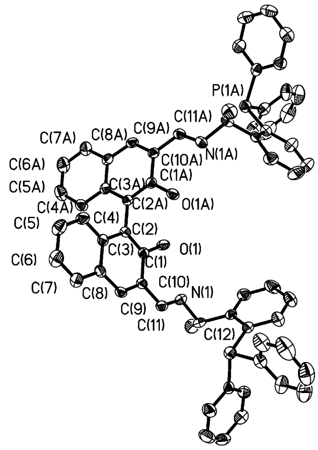
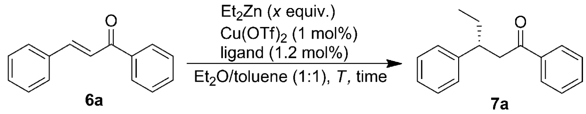


Figure 2. Crystal structure of chiral ligand (*R*,*S*,*S*)-**L1**.

Copper-Catalyzed Conjugate Addition of Et_2Zn to Enones

On the basis of the hypothesis of multinuclear bimetallic catalysis induced by a chiral source that bears both axial and sp^3-central chirality, we then examined these ligands in the model copper-catalyzed asymmetric conjugate addition of diethylzinc to a chalcone (Scheme 4 and Table 1). To



Scheme 4. A model conjugate addition of diethylzinc to chalcone for the screening of ligands.

Table 1. Screening of reaction conditions for copper-catalyzed conjugate addition of organozinc reagents to chalcone.[a]

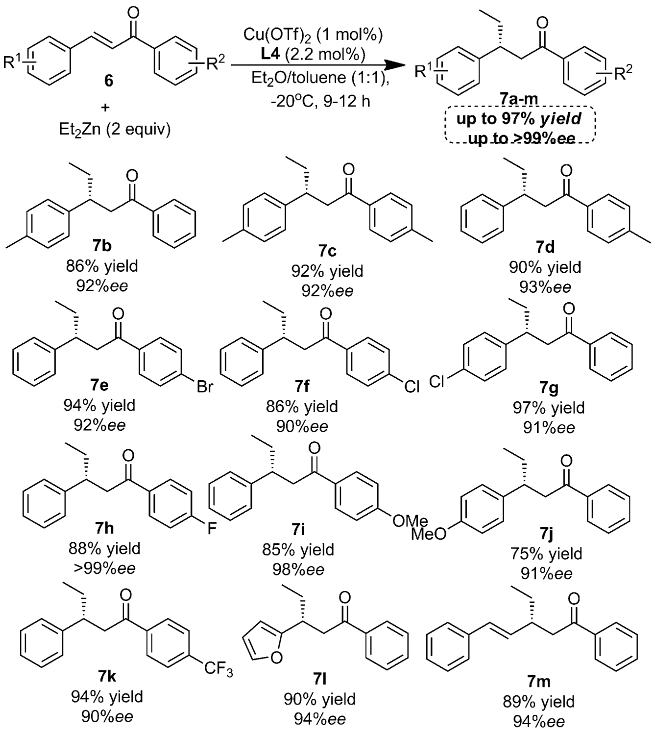
Entry	Ligand	*n*	*T* [°C]	*t* [h]	Yield [%][b]	*ee* [%][c]
1	**L1**	1.2	−20	14	40	84
2	**L2**	1.2	−20	12	15	−25
3	**L3**	1.2	−20	16	trace	nd
4	**L4**	1.2	−20	12	28	94
5	**L5**	1.2	−20	12	25	−37
6	**L6**	1.2	−20	12	70	71
7[d]	**L1**	1.2	−20	12	60	79
8	**L1**	1.2	−20	48	25	87
9[e]	**L1**	1.2	−20	12	25	77
10	**L1**	3.0	−20	12	75	83
11	**L1**	3.0	0	12	35	84
12	**L4**	3.0	−20	10	92	94
13[f]	**L4**	3.0	−20	10	85	91
14[g]	**L4**	3.0	−20	10	85	94
15[h]	**L4**	3.0	−20	10	80	50
16[i]	**L4**	3.0	−20	10	80	82
17	(*S*)-**1**	3.0	−20	16	40	14

[a] Reaction conditions: chalcone (1 mmol), $Cu(OTf)_2$ (1 mol %), chiral ligand (**L1**–**L3**: 1.2 mol %; **L4**–**L6**: 2.2 mol %). [b] Isolated yields. [c] The enantiomeric excess (*ee*) was determined by HPLC analysis on a chiral stationary phase (see the Supporting Information for details); nd = not determined. [d] 0.6 mol % of (*R*,*S*,*S*)-**L1** was employed. [e] 1,4-Dioxane was employed as a solvent in this case. [f] Toluene was employed as a solvent. [g] $CuCl_2$ was employed as a catalyst. [h] THF was employed as a solvent. [i] CH_2Cl_2 was employed as a solvent.

identify the optimal ligand structure and metal system, a series of copper salts in combination with chiral ligands was evaluated as enantioselective catalysts for the conjugate addition of Et_2Zn to chalcone **6a**. These experiments led to the determination of $Cu(OTf)_2$ (OTf = trifluoromethanesulfonate) as the best copper source in this reaction.[16] And then the six phosphine ligands that involved a Schiff base with the illustrated absolute configuration were investigated to confirm various levels of enantioselection. The representative results from the ligand survey with $Cu(OTf)_2$ are summarized in Table 1 (entries 1–6). From this survey and comparison, the novel BINOL-modified phosphine **L1** that contains the Schiff base derived from (*R*)-BINOL and (*S*)-1-phenylethanamine emerged as a promising multifunctional chiral ligand (84% *ee*). Notably and interestingly, ligand (*S,S,S*)-**4a** (**L2**) and ligand (*S,S*)-**4c** (**L5**) derived from (*S*)-BINOL and (*S*)-1-phenylethanamine exhibited poor activity in enantiomeric induction; they were incapable of achieving good yield and high enantioselectivity (only 25 or 37% *ee*, respectively; Table 1, entries 2 and 5). This result showed that the stereochemistry of the BINOL backbone played a crucial role in this copper-catalyzed conjugate addition. Moreover, the mismatch between different chiral sources, C_2-axial binaphthol and sp^3-central chiral phosphine, revealed that the geometric orientation of BINOL-derived phosphine was crucial to the enantioselective induction in this reaction. It should be noted that both the phosphine and hydroxyl moiety also proved to be very important in this reaction because almost no desired product was detected in the presence of phosphine-group-free ligand **L3** (Table 1, entry 3) and only moderate enantioselectivity and yield was obtained with Ar-BINMOL-derived ligand **L6** under the same reaction conditions (Table 1, entry 6).

Although the enantioselective induction of hexadentate ligand **L1** was not good in this reaction, further optimization of this process showed that the reaction might be performed with a lower amount of ligand **L1** because 0.6 mol% of **L1** (Cu/**L1** = 1.7:1) was found to be enough to give better yields (60%) and good enantioselectivity (Table 1, entry 7, 79% *ee*). Interestingly, a large amount of unidentified byproducts was detected when the reaction was prolonged to 48 h (Table 1, entry 8). Other solvents were also not suitable media for this reaction; for example, low yield and decreased enantioselectivity of the desired product **7a** was obtained in 1,4-dioxane (77% *ee*, 25% yield; Table 1, entry 9). Increasing the amount of Et_2Zn to 3 equiv led to the improvement of the isolated yield (75%; Table 1, entry 10), whereas with 1.2 equiv of Et_2Zn it was found to proceed smoothly but resulted in poor isolated yield because of a large amount of unidentified byproducts. Interestingly, increasing the reaction temperature to 0 °C also gave poor yield (35%) with no change in enantioselectivity (Table 1, entry 11). These results showed that hexadentate ligand **L1** was not suitable for the copper-catalyzed conjugate addition of diethylzinc to chalcone in terms of enantioselectivity (up to 87% *ee*) and moderate yield (up to 75% isolated yield). On a more positive note, the tetradentate ligand **L4** gave excellent enantioselectivity (Table 1, entries 4 and 12; 94% *ee*) in the presence of 1.2 or 3 equiv Et_2Zn, in which the latter conditions led to better isolated yield (92%; Table 1, entry 12). This work suggests that monophosphine **L4** is a promising ligand in this reaction. Subsequent study of the effect of different solvents showed that the enantioselectivity was also sensitive to the solvent. On the basis of these results, with the tetradentate ligand **L4**, we employed $Cu(OTf)_2$ and Et_2O at low temperature (−20 °C) in the following studies of substrate scopes, and some other choices, such as ligand **L1**, might also be effective.

With the optimized conditions and tetradentate ligand **L4** in hand, the scope of the copper-catalyzed conjugate addition reaction of various chalcones was explored (Scheme 5).

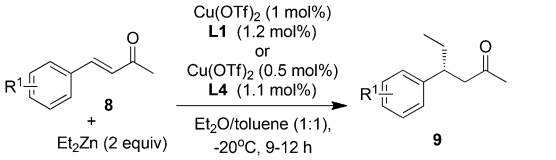


Scheme 5. Asymmetric Cu/**L4**-catalyzed conjugate addition of a diethyl zinc reagent to various chalcones.

In general, excellent enantiomeric excesses (90–98% *ee*) and good-to-excellent isolated yields (75–97%) were observed for aromatic chalcones **6a–m** that bear either electron-withdrawing or electron-donating groups. Furyl and styryl substituents did not show any adverse effect on the conjugate addition (94% *ee* for **7l** and **7m**). As summarized in Scheme 5, the present asymmetric copper–**L4** catalyst system was synthetically useful and applicable to various chalcones because all reactions were performed with only 1 mol% of copper and 2.2 mol% of chiral ligand **L4**.

The attempt to expand substrate scopes of various enones in the copper-catalyzed conjugate addition was also carried out in the presence of hexadentate ligand **L1** and tetradentate ligand **L4**. Fortunately, the substrate scope of the pres-

Table 2. Copper-catalyzed conjugate addition of Et_2Zn to 4-phenylbutenone and its analogues (**8**).

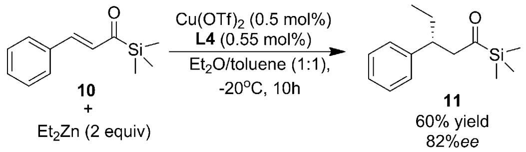


Entry	R[1]	Conditions[a]	Yield [%][b]	*ee* [%][c]
1	H	A	**9a**: 85	98
2	H	B	**9a**: 90	>99
3	*m*-OMe	B	**9b**: 80	97
4	*p*-OMe	A	**9c**: 82	99
5	*p*-OMe	B	**9c**: 97	>99
6	*o*-OMe	A	**9d**: 98	97
7	*o*-OMe	B	**9d**: 95	99
8	*m*-Br	A	**9e**: 94	97
9	*m*-Br	B	**9e**: 86	98
10	*p*-Br	A	**9f**: 90	97
11	*p*-Br	B	**9f**: 99	98
12	*p*-Cl	A	**9g**: 99	97
13	*p*-Cl	B	**9g**: 81	98
14	*p*-CF_3	B	**9h**: 96	>99
15[d]	2-naphthyl	A	**9i**: 99	97
16[d]	2-naphthyl	B	**9i**: 90	99
17	*p*-Ph	B	**9j**: 80	98

[a] Reaction conditions A: enone **8** (1 mmol), $Cu(OTf)_2$ (1 mol%), chiral ligand **L1** (1.2 mol%); conditions B: enone **8** (1 mmol), $Cu(OTf)_2$ (0.5 mol%), chiral ligand **L5** (1.1 mol%). [b] Isolated yields. [c] The enantiomeric excess (*ee*) was determined by HPLC analysis on a chiral stationary phase. [d] The enone is (*E*)-4-(naphthalen-2-yl)but-3-en-2-one.

ent catalyst system has proven to be very wide, and the best enantioselectivity was achieved with an almost optically pure form (99% *ee*) with 1 mol% of copper and 1.2 mol% of **L1** (conditions A) or with 0.5 mol% of copper and 1.1 mol% of **L4** (conditions B). As shown in Table 2, different from the above findings, both ligand **L1** and **L4** resulted in superior enantioselectivities ($\geq$97% *ee*) and excellent isolated yields (up to 99%) in the conjugate addition of 4-phenylbutenone and its analogues (**8**). Thus, the present hexadentate ligand **L1**-derived catalyst system has a potential substrate-selective characteristic to discriminate between the reactivity of acyclic enones and chalcones in terms of enantioselectivity. Particularly with the use of tetradentate ligand **L4** for the copper-catalyzed conjugate addition, most of the substrates resulted in almost perfect enantioselectvities (99% *ee*), and the electronic and steric properties of β-aryl substituents did not affect the selectivity. Clearly, for these substrates shown in Table 2, the catalytic performance in terms of enantioselectivity achieved by the use of ligand **L1** or **L4** has reached a new level relative to that of previous systems with copper catalysis.[17]

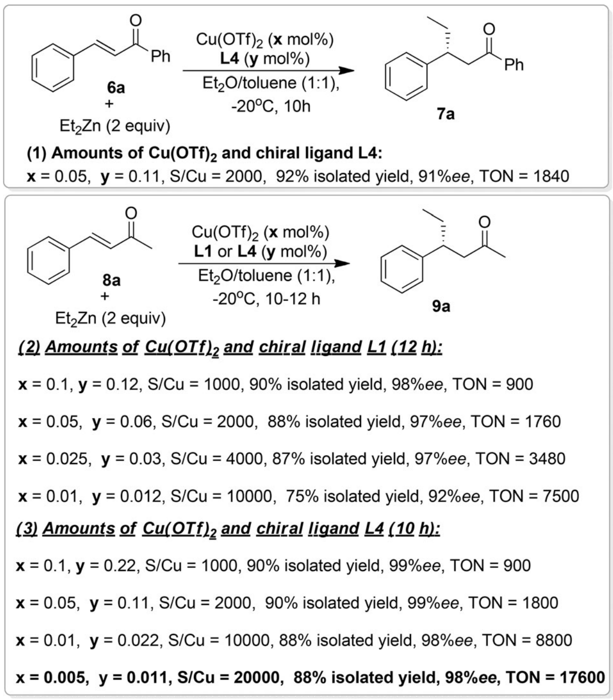
Although acylsilane is well documented to be an unstable compound relative to that of corresponding ketones with a similar structure,[18] the fact that α,β-unsaturated acylsilane was used as a special enone is interesting and worthy of investigation. Herein, we found that the (*E*)-3-phenyl-1-(trimethylsilyl)prop-2-en-1-one (**10**) was reactive with diethylzinc in the presence of ligand **L4**, in which the reaction gave



Scheme 6. Copper-catalyzed asymmetric conjugate addition of diethylzinc to α,β-unsaturated acylsilane.

the corresponding chiral acylsilane **11** in good enantioselectivity (82% *ee*) and moderate isolated yield (Scheme 6, 60% isolated yield).

On the basis of the excellent catalytic efficiency of multidentate phosphine ligands in these experimental results, we investigated the conjugate addition of Et_2Zn to chalcone (**6a**) or 4-phenylbutenone (**8a**) with low catalyst loading. The results of the attempts to reduce catalyst loading are summarized in Scheme 7. Initially, we found that 0.05 mol%



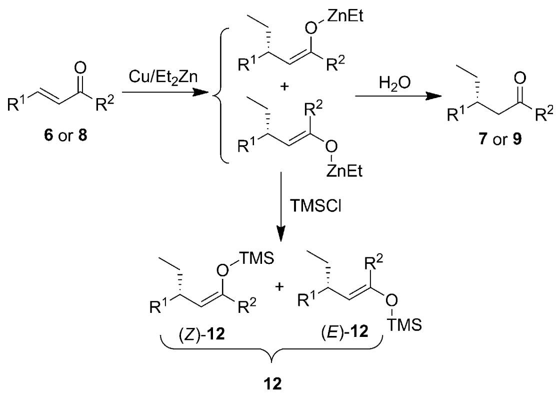
Scheme 7. A model example of highly efficient and enantioselective conjugate addition of $ZnEt_2$ to 4-phenylbutenone.

of copper catalyst was good enough in terms of enantioselectivity and yield for the desired product **7a** in the presence of tetradentate ligand **L4**. Surprisingly, when 4-phenylbut-3-en-2-one was used as a model substrate in the copper-catalyzed conjugate addition of diethylzinc reagent, even with the hexadentate ligand **L1**, the reaction could also proceed smoothly in the presence of 0.025 mol% of copper–**L1** complex (S/C = 4000). As shown in Scheme 7, the conjugate addition reactions were completed within 12 h when using 0.025–0.1 mol% catalyst to afford product **9a** in good isolated yields and excellent enantioselectivities (TON = 900–

3480, 87–90% isolated yields, 97–98% *ee*). Importantly, the conversion and enantioselectivity remained high when the reaction was performed with 0.01 mol% of catalyst (S/C = 10000), which gave the gram-scale product in notable 75% yield (isolated) and 92% *ee* (TON = 7500) with little inhibition of the catalytic activity. More importantly, the tetradentate ligand **L4** exhibited higher catalytic efficiency in this reaction. To compare the catalytic activity of **L4**, we chose 0.005 mol% of $Cu(OTf)_2$ as a model. It is clear that ligand **L4** is highly effective in this reaction, because 98% *ee* and 88% yield of the desired product **9a** was achieved under these conditions (Scheme 7; the highest TON is 17600). We believe that the amount of catalyst system (Cu–**L4**) could be reduced to some extent. Therefore, we have succeeded in the development of a highly efficient catalytic system for the conjugate addition of alkyl zinc reagent to acylic enones with the best catalytic performance to date.[19]

Trapping of the Zinc Enolates: Conjugate Addition–Silylation and Conjugate Addition–Acylation

The common feature of asymmetric copper-catalyzed conjugate addition of organozinc reagents is the generation of enantiomerically enriched zinc enolates. Thus trapping of the zinc enolate with an electrophile could be a conjugate-addition-initiated tandem transformation, which would quickly lead to more complex molecules.[20] In light of the importance of silyl enol ethers for organic synthesis,[21] it would be desirable to prepare them by trapping zinc enolates with a silylating reagent in situ (Scheme 8). In addition,

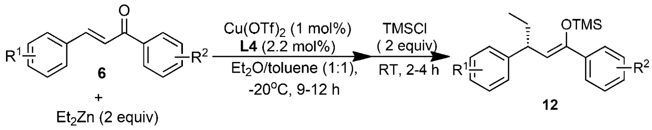


Scheme 8. Trapping of zinc enolates with water and chlorotrimethylsilane (TMSCl).

the selective silylation of zinc enolate intermediates might also shed some light on the mechanism of copper-catalyzed conjugate addition of diethylzinc reagent in the presence of multidentate phosphine ligand **L1** or **L4**.

Inspired by previous findings of enantioselective copper-catalyzed conjugate addition–silylation of zinc enolates,[22] we studied the tandem silylation of enantiomerically enriched zinc enolates that formed in situ from copper–**L4** complex-catalyzed conjugate addition. In the present copper-catalyzed conjugate addition–silylation, we first investigated the conjugate-addition-initiated silylation of chalcone **6a** by using 1 mol% $Cu(OTf)_2$ under optimal conditions with tetradentate ligand **L4**. The reaction gave the corresponding product **12a** with excellent enantioselectivity, high *Z* selectivity, and good yield (Table 3, entry 1). Thus it

Table 3. Copper-catalyzed conjugate addition–silylation.[a]



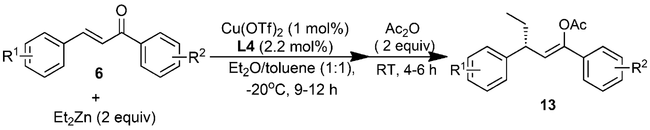
Entry	R^1	R^1	Yield [%][b]	*ee* [%][c]	*Z*/*E*[d]
1	H	H	**12a**: 75	94	>99:1
2	*p*-Me	H	**12b**: 82	92	>99:1
3	*p*-Me	*p*-Me	**12c**: 83	92	>99:1
4	H	*p*-Me	**12d**: 83	93	>99:1
5	H	*p*-Br	**12e**: 70	92	>99:1
6	H	*p*-Cl	**12f**: 85	90	>99:1
7	*p*-Cl	H	**12g**: 83	91	>99:1
8	H	*p*-F	**12h**: 73	>99	>99:1
9	H	*p*-CF_3	**12i**: 75	91	>99:1
10	*p*-CF_3	H	**12j**: 78	90	>99:1

[a] Reaction conditions: enone **6** (1 mmol), $Cu(OTf)_2$ (1 mol%), chiral ligand **L4** (2.2 mol%), TMSCl (2 equiv). [b] Isolated yields. [c] The enantiomeric excess (*ee*) after desilylation to the corresponding ketone was determined by HPLC analysis on a chiral stationary phase. [d] The ratio of *Z*/*E* was determined by GC-MS and ^{1}H NMR spectroscopy.

is clear that a (*Z*)-zinc enolate intermediate was formed in this conjugate addition of diethylzinc to chalcone. As previously observed, substituted groups on the aryl ring almost have no negative effect on the tandem conjugate addition–silylation reaction. Whereas we used in situ formed zinc enolates and chlorotrimethylsilane for the initial development of conjugate addition–silylation, purification of these silyl enol ethers was a nontrivial task due to the hydrolysis of the desired compounds on chromatography supports. As shown in Table 3, the results of this reaction suggest that various air-stable and enantiomerically enriched (*Z*)-silyl enol ethers could be easily prepared by the tandem copper-catalyzed conjugate addition–silylation of chalcones.

As a potential alternative to silylation trapping of zinc enolates, tandem conjugate addition–acylation has been demonstrated successfully in recent years.[23] The formation of a more stable enolacetate derived from enantiomerically enriched zinc enolates would thus be a special way to prepare chiral esters. We therefore continued to study the possibility of acylation for the trapping of zinc enolates. With the conditions developed above in hand, we next investigated the acylation of zinc enolates. As expected, similarly to the copper-catalyzed conjugate–silylation reaction, the use of acetic anhydride as an alternative electrophile to trap zinc enolate led to the high-yield preparation of enoacetates. As shown in Table 4, this methodology could be applied to various chalcones. For all the substrates investigated, the chiral enoacetates were formed in good yields over the two-step addition/acylation sequence. After recognizing the ef-

Table 4. Copper-catalyzed conjugate addition-acylation.[a]



Entry	R^1	R^1	Yield [%][b]	*ee* [%][c]	*Z*/*E*[d]
1	H	H	**13a**: 80	94	>99:1
2	*p*-Me	H	**13b**: 80	92	>99:1
3	*p*-Me	*p*-Me	**13c**: 83	92	>99:1
4	H	*p*-Me	**13d**: 77	93	>99:1
5	H	*p*-Br	**13e**: 75	92	>99:1
6	H	*p*-Cl	**13f**: 82	90	>99:1
7	*p*-Cl	H	**13g**: 80	91	>99:1
8	H	*p*-F	**13h**: 82	>99	>99:1
9	H	*p*-CF_3	**13i**: 77	91	>99:1
10	*p*-CF_3	H	**13j**: 81	90	>99:1
11	Styryl	H	**13k**: 85	94	>99:1

[a] Reaction conditions: enone **6** (1 mmol), $Cu(OTf)_2$ (1 mol%), chiral ligand **L4** (2.2 mol%), Ac_2O (2 equiv). [b] Isolated yields. [c] The enantiomeric excess (*ee*) after deacylation to the corresponding ketone was determined by HPLC analysis on a chiral stationary phase. [d] The ratio of *Z*/*E* was determined by GC-MS and ^{1}H NMR spectroscopy.

fectiveness of copper-catalyzed conjugate addition–silylation and conjugate addition–acylation, we anticipated that the present pathway to trap zinc enolates would feature mild conditions, good yield, low catalyst loading, and high diastereoselectivity, as well as the inherently high level of enantioselectivity of asymmetric conjugate addition.

Mechanistic Study of Copper-Based Bimetallic Catalyst Systems

Although it is premature to discuss details of the mechanism of the whole catalytic cycle induced by the copper or bimetallic copper–zinc catalytic system, it is necessary to reveal the specialized structure of the multifunctional phosphine ligand **L1** or **L4** and the corresponding copper catalyst system that involves zinc- or titanium-based coordination. Being aware of the X-ray structure of the chiral ligand **L1** shown in Figure 1, we assumed that the protracted geometry with Schiff base linked chiral phosphine oriented toward the vacant apical positions would be beneficial to the construction of bimetallic multinuclear or polymetallic complex derived from copper and titanium or zinc.

On the basis of the above reaction results, we hypothesized that the intermolecular interaction and coordination of Schiff base–phosphine ligand **L1** or **L4** with copper in the presence of the second metal (Zn or Ti) led to the in situ formation of a polymer-like metal-based network (Figure 3). For hexadentate ligand **L4**, it is reasonable to deduce that the bimetallic complex with a large cavity or hole in this polymetallic complex exhibited promising size-selective activation among aromatic ketone-derived chalcones (**6**) and acetone-derived enones (**8**) as well as discrimination between enantioselectivity and reactivity during the addition of Et_2Zn to enones.

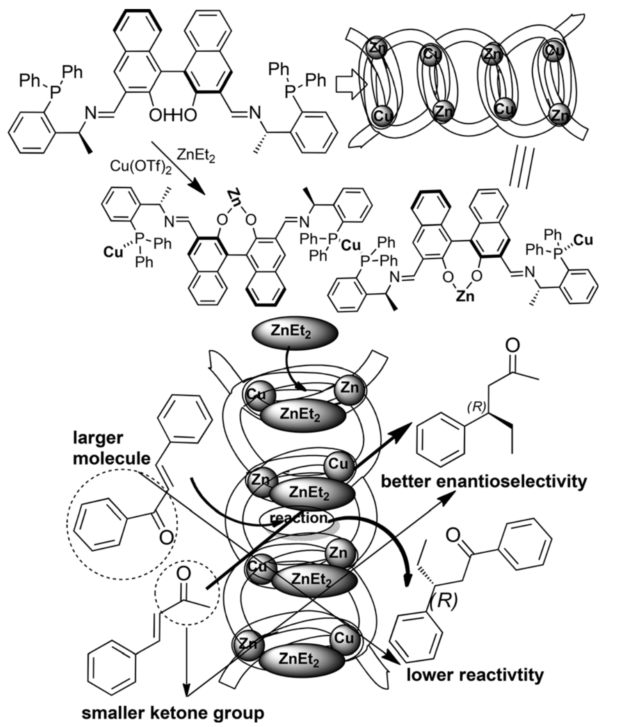
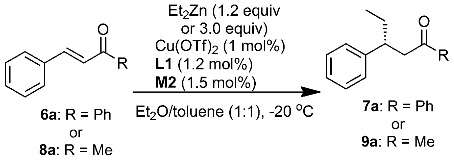


Figure 3. Schematic view of the self-assembled construction of a multinuclear polymetallic complex of **L1**.

It would be interesting to confirm the formation of a copper- and zinc-based polymetallic complex with hexadentate phosphine ligand **L1**. General spectral analysis would be useful for the elucidation of the present catalyst system. However, the instability of diethylzinc in air made it difficult to recover the polymetallic catalyst after reaction. With such promising findings in copper-catalyzed conjugate addition of the diethylzinc reagent to chalcone, we proposed that hexadentate phosphine ligand **L1** could work with both copper and zinc as a multinuclear bimetallic complex to activate chalcone in this reaction because the zinc metal linked with two hydroxyl groups in the BINOL-derived phosphine was important for achieving excellent catalytic performance. Therefore the effect of a second metal linked with a diol moiety on the enantioselectivity of copper-catalyzed conjugate addition of Et_2Zn to chalcone **6a** in the presence of hexadentate phosphine ligand **L1** would be important. In addition, although the enantioselective activity of **L1** is not good, we expected that the impact of a second metal salt on enhancing the chirality or weakening the enantioselectivity and reactivity would also supplement our structural elucidation of the catalyst system.

Initially, on the basis of the hypothesis that the ligation of zinc to ligand would be crucial to the excellent enantioselectivity and reactivity, we examined the effectiveness of the second metal catalyst instead of zinc in the copper-catalyzed asymmetric conjugate addition of organozinc reagents. In the first screening, we investigated $Ti(OiPr)_4$, $Al(OiPr)_3$, $InBr_3$, $Ni(OAc)_2$, $KOtBu$, and $Si(OEt)_4$, which could be treated with the BINOL moiety to form a BINOLate metal

Table 5. The effect of the second metal catalyst on copper-catalyzed conjugate addition of Et_2Zn to chalcone.[a]



Entry	*x* equiv Et_2Zn	R	**M2** [1.5 mol %]	*t* [h]	Yield [%][b]	*ee* [%][c]
1	1.2	Ph	–	14	40	84
2	1.2	Ph	Ti(O*i*Pr)$_4$	18	70	83
3	1.2	Ph	Al(O*i*Pr)$_3$	16	50	53
4	1.2	Ph	$InBr_3$	48	trace	–
5	1.2	Ph	$Ni(OAc)_2$	12	50	55
6	1.2	Ph	$Si(OEt)_4$	18	trace	–
7	3.0	Ph	–	12	75	83
8	3.0	Ph	Ti(O*i*Pr)$_4$	12	90	87
9	3.0	Ph	KO*t*Bu	12	88	55
10	3.0	Me	Ti(O*i*Pr)$_4$	12	85	97
11	3.0	Me	–	12	85	98

[a] Reaction conditions: chalcone (1 mmol), $Cu(OTf)_2$ (1 mol %), the second metal compound (**M2**, 1.5 mol %), chiral ligand **L1** (1.2 mol %). [b] Isolated yields. [c] The enantiomeric excess (*ee*) was determined by HPLC analysis on a chiral stationary phase (see the Supporting Information for details).

complex[24] as cocatalyst in this copper-catalyzed conjugate addition of Et_2Zn to chalcone **6a**. As shown in Table 5, these metal compounds had a promising effect on the conversion and enantioselectivity. The results in Table 5 as well as Table 1 also supported the notion that the combined use of copper and zinc or titanium resulted in a novel multinuclear catalyst system. The superior levels of reaction efficiency observed with 1.5 mol % of Ti(O*i*Pr)$_4$ in the presence of 1.2 equiv Et_2Zn (70 versus 40 % yield; Table 5, entries 1 and 2) or 3.0 equiv Et_2Zn (90 versus 75 % yield and 87 versus 83 % *ee*; Table 5, entries 7 and 8), prompted us to consider that the ligation of Ti with the BINOL backbone led to a better polymetallic catalyst system. Notably, as shown in entries 10 and 11 of Table 5, the use of Ti(O*i*Pr)$_4$ still retained excellent enantioselectivity (97 % *ee*) relative to that of the titanium-free reaction conditions. During the experiments, we observed that the titanium-based copper-catalyst system was also found to be effective in the conjugate addition of various chalcones (up to 93 % *ee*; see the Supporting Information), which also provided an efficient procedure to synthesize chiral ketones catalyzed by a Cu/Zn/Ti-based multimetallic catalyst.

On the basis of these control experiments and the facts summarized in Table 5 for the bimetallic conjugate addition of diethylzinc to chalcone, it seems reasonable that the ligand-linked bimetallic catalyst system did indeed exhibit a dramatic acceleration of the conjugate addition. However, it is difficult to explain why the reactivity is largely different between chalcones **6** and enones **8** in the presence of hexadentate phosphine ligand **L1**. The substrate-sensitive copper/zinc-based bimetallic complex might result in a rigid polymetallic material of a certain size (Figure 3), in which the chalcone molecule in its extended conformation does not fit into this rigid coordination environment completely. The large difference between enones **8** and **10** can be rationalized on the basis of the position of the methyl group and bulky trimethylsilyl group that control the relative orientation between the polymetallic catalyst system and substrates. Consistent with the schematic view in Figure 3, the self-assembled construction of the possible polymer-like multinuclear bimetallic complexes with a particular cavity is featured along with substrate sensitivity in the conjugate addition of organozinc reagents to enones, in which the catalytic behavior is similar to the metal–organic framework (MOF)-based catalyst.[25,26]

To gain information on the structure of the copper and ligand **L1**-based polymetallic complex, we initially carried out a UV spectra and circular dichroism (CD) analysis of the chiral ligand **L1** and the metal (Cu, Ti, Cu–Ti) complexes, respectively. Although we were unsuccessful in obtaining the X-ray structure of the exact copper/zinc- or copper/titanium-based multinuclear metal complexes, circular dichroism (CD) was an especially powerful method to study the orientation of the molecules within the metal complex. It has been reported to be a powerful analytical tool for the study of supramolecular recognition or complicated metal-based macromolecular catalyst systems.[27]

The UV spectra and circular dichroism (CD) spectra of the chiral ligand **L1** and the metal (Cu, Ti, Cu–Ti) complexes are provided in Figure 4 (also see the Supporting Information). As expected, coordination of the chelating hexadentate phosphine ligand **L1** to Cu, Ti, or bimetallic Cu–Ti caused considerable changes in the CD spectra (Figure 4). The CD spectra of ligand **L1** exhibit two major signature bands that correspond to naphthyl π–π* transitions at 225 and 257 nm for **L1**. For the bimetallic complex derived from

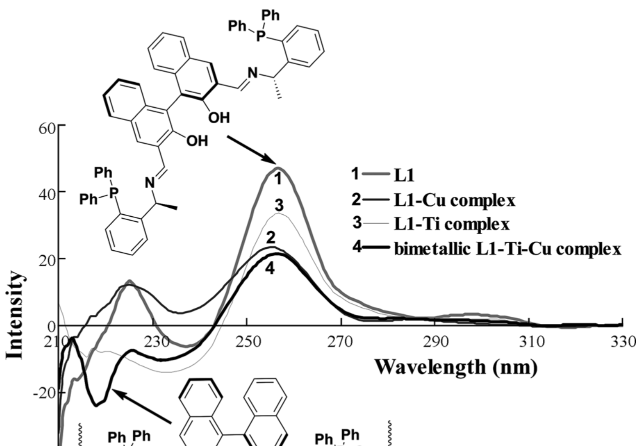


Figure 4. CD spectra of chiral ligand (*R*,*S*,*S*)-**L1** and the mixture of (*R*,*S*,*S*)-**L1** with Ti(O*i*Pr)$_4$, $Cu(OTf)_2$, and the combined use of Ti(O*i*Pr)$_4$ and $Cu(OTf)_2$ in CH_2Cl_2.

the combined use of titanium and copper, it allows for the construction of a Ti-Cu-**L1**-based polymetallic complex upon guest coordination, thus resulting in larger geometry changes for the chirality moiety and giving rise to larger changes in the CD spectra (210–250 nm; Figure 4) due to phenyl π–π* transitions at 218 nm. The CD spectra of the copper−**L1** complex and titanium−**L1** complex exhibited different signature bands at 220–238 nm owing to the different coordination or interaction in which titanium is linked with the oxygen atom of the BINOL backbone and copper is coordinated with the phosphine or nitrogen atom of the Schiff base moiety. Similarly, the CD spectra of the chiral ligand **L4** and the metal (Cu, Ti, Cu−Ti) complexes (Figure 5) also

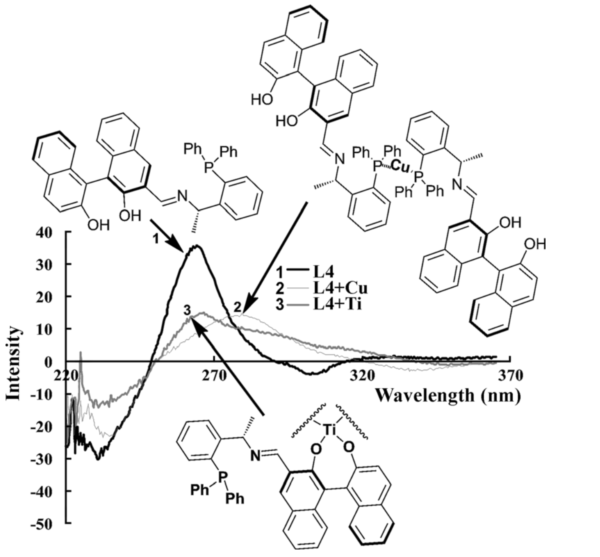


Figure 5. CD spectra of chiral ligand (*R*,*S*)-**L4** and the mixture of (*R*,*S*)-**L4** with Ti(O*i*Pr)$_4$ or Cu(OTf)$_2$ in CH_2Cl_2.

supported the notion that the coordination of copper and phosphine resulted in notable changes in the CD spectra. Specifically, the most intense band maxima shift by about 16 nm (from 264 to 280 nm). Therefore, when it comes to the catalytic asymmetric conjugate reaction with the copper−**L4** or −**L1** system, we propose that the observed reaction that results in this conjugate addition might be attributed to the effect of the chiral ligand **L1** or **L4** that contains both axial and sp^3-central chirality oriented toward the assembled formation of an ordered copper-linked polymetallic complex. We believe that the steric hindrance of the chiral Schiff base around the dihydroxy groups in hexadentate ligand **L1** is responsible for its lack of reactivity with diethylzinc, which led to the decrease of enantioselectivity and reactivity relative to that of tetradentate ligand **L4**.

Another attempt to elucidate the Ti-Cu-**L1**-based polymetallic complex by ESI-MS and MALDI-TOF MS analysis was unsuccessful at present owing to the probably high molecular weight, and the possible interpenetrating network of this complex. We then performed an ESI(+)-MS analysis of the mixture of copper and tetradentate **L4** under the same

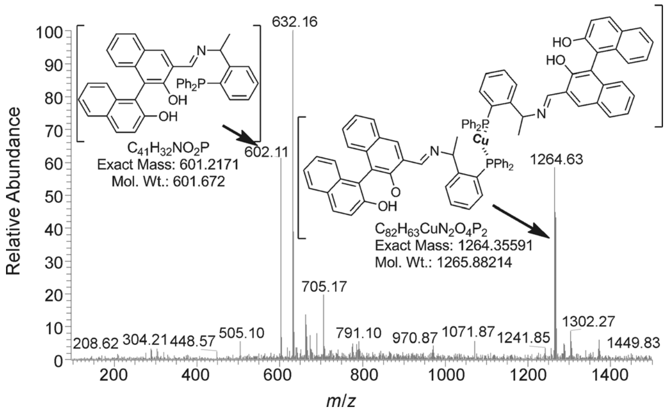


Figure 6. ESI(+)-MS for the mixture of Cu(OTf)$_2$ with phosphine ligand **L4**.

conditions. Figure 6 shows that the ESI(+)-MS data identified the chiral copper-**L4** complex intermediate of *m*/*z* 1264.63. In another regard, the negative results of the MS analysis of copper−**L1** also supported the existence of the possible polymetallic complex (Figure 3) but not a simple small-molecule copper complex or dimeric complex.

The possibility of a polymetallic complex was also confirmed by ^{31}P NMR spectroscopy (Figure 7). ^{31}P NMR spec-

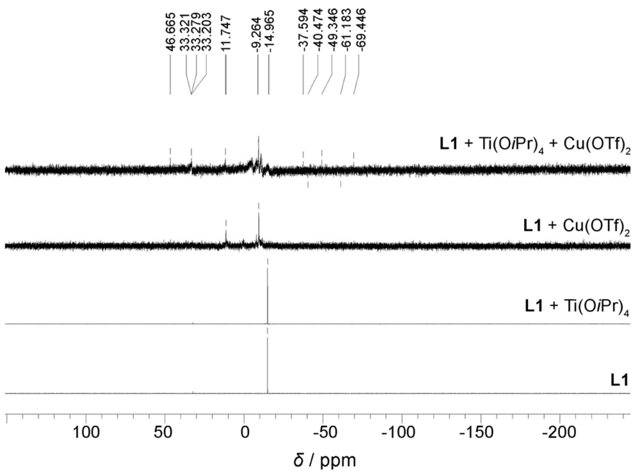


Figure 7. ^{31}P NMR spectra of chiral ligand (*R*,*S*,*S*)-**L1** and the mixture of (*R*,*S*,*S*)-**L1** with Ti(O*i*Pr)$_4$, Cu(OTf)$_2$, and the combined use of Ti(O*i*Pr)$_4$ and Cu(OTf)$_2$.

troscopy revealed no interaction between titanium and the phosphine center, although there was strong and competitive coordination of copper with nitrogen and the phosphine center. The ^{31}P NMR spectra of the Ti/Cu-based catalyst system showed unordered signals, thereby implying that the Ti-Cu-**L1** complex has a complicated structure due to the possibility of polymetallic coordination with the phosphine ligand. Thus, similarly to previous reports,[3–8] the novel multifunctional chiral phosphine was formed through noncovalent-interaction-driven assembly as well as O−M (Zn or Ti) bond-forming coordination.

Conclusion

In summary, we have successfully synthesized a new family of chiral Schiff base–phosphine ligands derived from chiral binaphthol (BINOL) and chiral primary amine. The controllable synthesis of a new family of multifunctional Schiff base–phosphine ligands (HZNU-Phos) that contain both axial and sp^3-central chirality from axial BINOL and sp^3-central primary amine led to the establishment of an efficient multifunctional phosphine ligand for copper-catalyzed conjugate addition of organozinc reagents. Thus, we have established a highly efficient and enantioselective catalyst system comprised of a bimetallic copper–zinc complex for conjugate addition of organic zinc reagents to aromatic enones. In the asymmetric conjugate reaction of organozinc reagents to enones, the polymer-like bimetallic multinuclear Cu−Zn complex constructed in situ was found to be a substrate-selective and excellent catalyst for the diethylzinc reagent in terms of enantioselectivity (up to >99% *ee*). More importantly, the match in chirality between different chiral sources, C_2-axial binaphthol and sp^3-central chiral phosphine, was crucial to the enantioselective induction in this reaction. The experimental results indicated that our chiral ligand (*R*,*S*,*S*)-**L1** and (*R*,*S*)-**L4**-based bimetallic complex catalyst system exhibited the highest catalytic performance to date in terms of enantioselectivity and conversion, even in the presence of 0.005 mol% of catalyst (S/C = 20000, TON = 17600). We also studied the tandem silylation or acylation of enantiomerically enriched zinc enolates that were formed in situ from copper−**L4**-complex-catalyzed conjugate addition, which resulted in the high-yield synthesis of chiral silyl enol ethers and enoacetates, respectively. Furthermore, the specialized structure of multifunctional phosphine ligand **L1** or **L4** was investigated and a corresponding mechanistic study of the copper-catalyst system was carried out by means of ^{31}P NMR spectroscopy, circular dichroism (CD), and UV/Vis absorption. On the basis of the above measurements, we suggest that the intermolecular interaction and coordination of Schiff base–phosphine ligand **L1** with copper in the presence of the second metal (Zn or Ti) led to the in situ formation of a polymer-like network and dynamic chiral multinuclear bimetallic complex, which resulted in highly efficient activation of enone and spectacular enantioselective induction. Thus it is an important complementary method to the asymmetric synthesis of various optically active β-ethyl ketones because a new level of catalytic performance was reached relative to the previously reported chiral phosphine ligand systems for acyclic enones, especially for 4-phenylbutenone and its analogues (with a high TON and up to 99% *ee*). Further investigations into the substrate-sensitive mechanism and the intricate structure of this polymetallic complex are currently underway and will be reported in due course.

Experimental Section

Reagents were purchased from commercial sources and were used as received unless noted otherwise. Reactions were monitored by thin-layer chromatography using silica gel. All the reactions that dealt with air- or moisture-sensitive compounds were carried out in a dry reaction vessel under positive pressure of argon. Air- and moisture-sensitive liquids and solutions were transferred with a syringe or a stainless-steel cannula. ^{1}H NMR spectra were recorded at 400 MHz; chemical shifts are reported in ppm relative to tetramethylsilane (TMS) with the solvent resonance employed as the internal standard ($CDCl_3$ at $\delta = 7.26$ ppm). ^{13}C NMR spectra were recorded at 100 MHz; chemical shifts are reported in ppm from TMS with the solvent resonance as the internal standard ($CDCl_3$ at $\delta = 77.20$ ppm). ^{31}P NMR spectra were recorded at 162 MHz; chemical shifts for phosphorous are reported in ppm referenced to the phosphorous resonance of phosphoric acid ($\delta = 0$ ppm). ESI-MS analysis of the samples was conducted using an LCQ Advantage mass spectrometer (ThermoFisher Company, USA) equipped with an ESI ion source in the positive-ionization mode, with data acquisition using the Xcalibur software (Version 1.4). CH_2Cl_2 was dried and distilled over CaH_2. THF and Et_2O were distilled from sodium benzophenone ketyl. Aldehydes and titanium tetraisopropoxide were used after distillation.

CCDC-913453 ((*R*,*S*,*S*)-**L1**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

X-ray Diffraction

Data sets were collected using Bruker APEX DUO and Bruker APEX-II CCD diffractometers. Programs used: For data collection Bruker APEX2,[28a] data reduction Bruker SAINT, absorption correction multi-scan, structure solution SHELX-97,[28b] structure refinement SHELXL-97,[28b] and for graphics Bruker SHELXTL.[28b] The detailed procedures for the synthesis of every chiral ligand and copper-catalyzed conjugate addition of organozincs to enones have been provided in the Supporting Information. In the copper-catalyzed conjugate addition of diethylzinc to enones, both method A and method B with different N,O,P ligands led to the same products (see the Supporting Information); their differences are only in terms of the enantiomeric excesses. These different reaction results have been collected in the product characterization.

Representative Procedure for Copper-Catalyzed Conjugate Addition of Et_2Zn to Enones

A flame-dried Schlenk tube was charged with $Cu(OTf)_2$ (3.6 mg, 0.01 mmol) and (*R*,*S*,*S*)-**L1** (11.0 mg, 0.012 mmol) under an N_2 atmosphere, and the mixture was dissolved in dry Et_2O (3.0 mL). The solution was stirred at room temperature for 30 min and then cooled to −20 °C. Diethylzinc (3.0 mmol, 3.0 mL of 1 M toluene solution) was added dropwise to the above solution. Enone (1.0 mmol) was added at once to the clear yellow solution. The mixture was stirred at −20 °C for 9–12 h before being quenched with aqueous saturated NH_4Cl. The layers were separated, and the aqueous layer was extracted with ethyl acetate (5 mL twice). The combined organic layers were dried over Na_2SO_4 and concentrated under reduced pressure. The residue was purified by column chromatography on silica gel to give the addition product. The enantiomeric excess of the product was determined by chiral HPLC. The detailed experimental procedures and characteristics of corresponding products are provided in the Supporting Information.

Acknowledgements

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